



# Long-term diatom response to anthropogenic stressors in a Neotropical lake

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## Abstract

Understanding tropical lake ecosystem responses to anthropogenic disturbance over long timescales is critical for assessing resistance, resilience, and regime shifts; however, such perspectives are rare in the Neotropics. We combined paleolimnological, limnological, and remote-sensing approaches to evaluate the response of a shallow tropical maar lake in the Sierra de Los Tuxtlas (eastern Mexico) to deforestation-driven disturbance over the last ~200 years, with emphasis on ecosystem trajectories following peak impact in the late twentieth century. Forest-cover changes were reconstructed from remote images (1976, 2000, 2024), while lake condition was assessed using limnological measurements and diatom assemblages from the water column (2000–2024) and sediments (~1808–2025). Sedimentary magnetic susceptibility and diatoms documented a major state shift during the 1970s–1980s, coincident with peak deforestation and erosion, characterized by high species turnover and the dominance of disturbance-tolerant taxa (*Achnanthes minutissimum*–*Fragilaria capucina*). This transition contrasts with gradual and elastic responses observed during earlier periods. Despite partial reforestation and landscape stabilization since the 1990s, limnological variables indicate persistent eutrophic to hypereutrophic conditions. However, sedimentary and water-column diatoms revealed ongoing ecological changes, marked by declines in disturbance taxa and increasing abundances of baseline planktonic species (*Aulacoseira* spp.), indicating an elastic recovery trajectory toward pre-disturbance conditions. Comparisons with an earlier disturbance between AD 100–800 (Classic Mesoamerican period) highlight the importance of disturbance duration and climatic context in determining ecosystem resilience. Our results demonstrate long-term resilience in a tropical lake following intense anthropogenic disturbance, underscoring the value of paleolimnological records for evaluating ecosystem recovery.

**Keywords** Deforestation · Human impact · Paleolimnology · Tropical lakes · Ecological resilience

## Introduction

One of the most relevant questions in ecology and environmental management is to understand how ecosystems respond to disturbance over time, whether they persist (resistance) or transition toward new regimes or states, and whether they eventually return to their baseline or reference conditions once the disturbance has ceased (resilience) (Lake 2013). These transitions can be gradual (linear, with low species turnover) or abrupt (threshold, with high species turnover), while elastic responses occur when the ecosystem recovers to its previous baseline condition (Lake 2013; Van Meerbeek et al. 2021). A basic step toward understanding these processes is to document the evolution of disturbed or restored ecosystems over a sufficiently long time interval. Most perturbation and/or restoration studies are limited by the timeframe of the observations that can

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be made, as continuous monitoring for relatively long periods of time (e.g., > 10 years) is resource consuming. Lakes represent valuable models for studying ecosystem perturbation because they are among the most sensitive ecosystems to human-induced disturbances and are isolated systems where changes can be relatively easily quantified (Spears et al. 2017; Whittaker et al. 2017). Furthermore, the analysis of lake sediments offers an alternative for long-term lake monitoring, as paleolimnological methods represent an exceptional tool for the reconstruction of lake characteristics over long periods of time, allowing for the identification of lake conditions prior to perturbation (reference conditions) and to follow up lake responses through time for periods that exceed most monitoring programs (Battarbee 2000; Smol 2009). This approach is particularly relevant in regions where lake monitoring is not an established practice or for time intervals that predate lake monitoring programs; for example, paleolimnological studies in Lake Ontario, the easternmost of the Laurentian Great Lakes at the Canada–USA border, and its long-term eutrophication history still to be identified (Pillsbury et al. 2021). Forest clearance and increased phosphorous loading to the lake caused eutrophication, with a shift from its baseline conditions during the 1920s, favoring high phosphorus tolerant taxa. This impacted condition lasted until the 1980s, when measures were implemented to reduce phosphorous loadings and the phytoplankton community partially recovered. Similarly, in Lake Peñasquito, a Neotropical, karstic lake in southern Mexico, paleolimnological studies allowed degradation and eutrophication processes associated with expanding agricultural activities during the 1970s to be identified. In this case, the degradation process accelerated in the early 2000s, leading to an abrupt change in the lake conditions, with a high species turnover that favored taxa tolerant to higher turbidity and nutrient levels (Caballero et al. 2025).

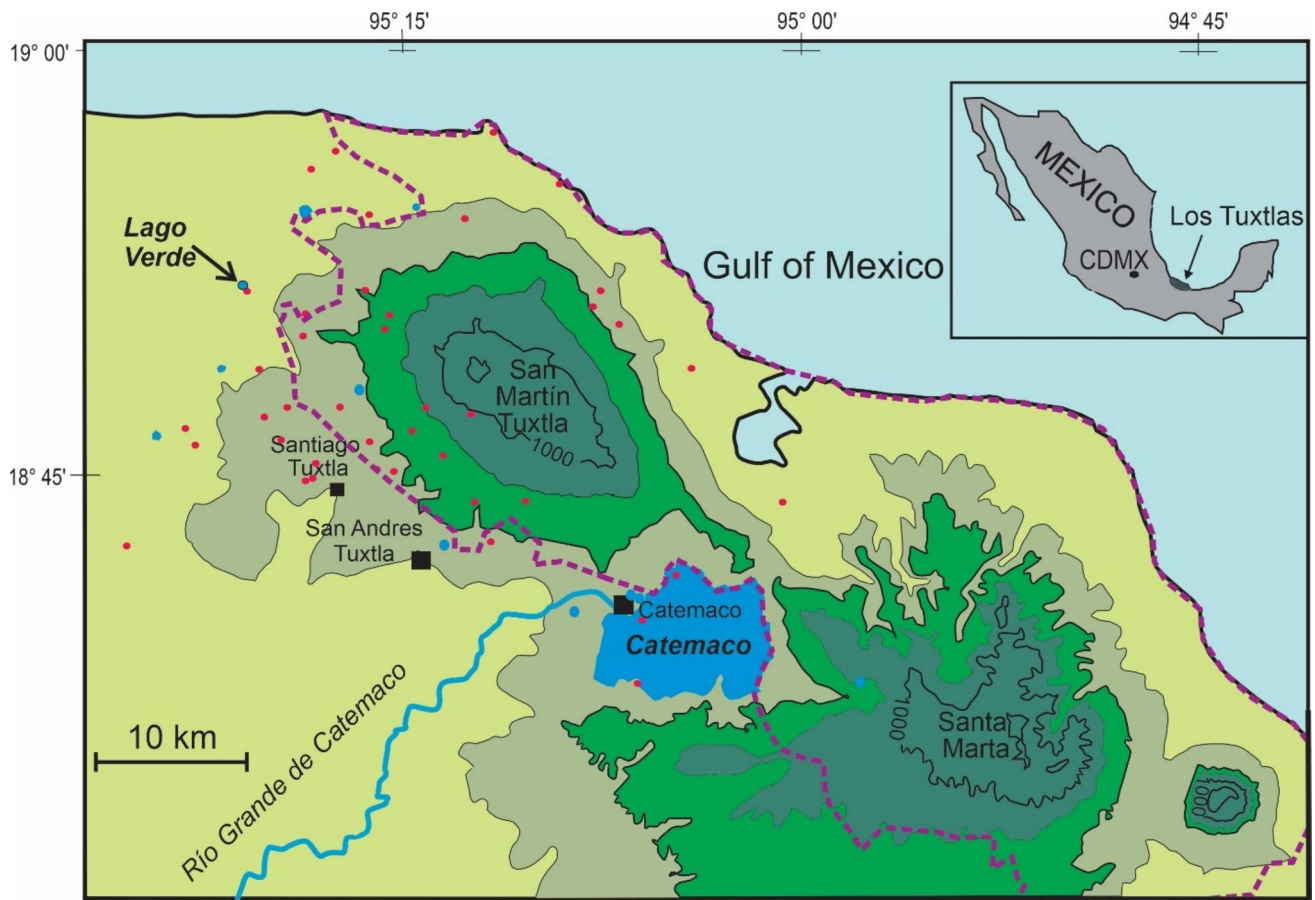
Neotropical ecosystems are some of the most biodiverse in the world; however, they have been severely affected by deforestation, particularly since the 1970s when regional economic development was frequently associated with forest clearing and conversion to pastures or farmlands (Bianchi and Haig 2013; Guevara and Laborde 2012; Rosa et al. 2016). The Sierra de los Tuxtlas (STUX) in eastern Mexico (Fig. 1) is a good example of how deforestation has altered Neotropical ecosystems. The STUX has been severely affected by deforestation and landscape fragmentation, losing about 40% of its forested area during the 1970s and 1980s (Guevara et al. 2004; Von Thaden et al. 2020). Even though deforested areas are frequently abandoned after a few years, allowing for reforestation processes, until the 1990s, deforestation in the region occurred at much higher rates than reforestation (Guevara et al. 2004). Due to its biological relevance, in 1998, a large part of the STUX was declared a protected natural reserve (Los Tuxtlas Biosphere

Reserve (Semarnat-Conanp 2006)), and some actions were implemented to reduce deforestation in the region, albeit with limited long-term success (Von Thaden et al. 2018).

Deforestation and erosion in the Neotropics impact aquatic ecosystems, as they lead to an increase in sediment influx, higher turbidity, and eutrophication (Callisto et al. 2014). In the STUX, many lakes have been impacted by deforestation and are now eutrophic to hypertrophic (Vázquez and Caballero 2013; Vázquez et al. 2004). However, the previous disturbance history of these lakes or their baseline or reference conditions before the modern deforestation process is mostly unknown. One exception is the maar lake Lago Verde, which offers a unique window into the long-term memory of disturbance in the Neotropics. In this lake, a paleolimnological study was conducted on a sediment core retrieved in 2001 (Caballero et al. 2006; Lozano-García et al. 2010; Ortega-Guerrero et al. 2006). This work showed a nearly 2000-year record of lake history in which two periods of disturbance associated with high erosion rates, deforestation, and eutrophication were identified; the first occurring during the Mesoamerican Classic period (AD 100 to 800) and the second associated with the modern deforestation process, which peaked during the 1970s and 1980s. In the early 1970s, the lake showed a sharp shift in diatom assemblages in response to high erosion and remained in this impacted condition until the top of the sequence, dating to 2001. Nearly 25 years later, we returned to Lago Verde with the objective of assessing its present condition to evaluate whether it has remained stable in its impacted condition, transitioned to a different regime, or returned to its previous reference condition. We hypothesized that landscape stabilization after peak disturbance and the establishment of the natural reserve in 1998 could have been associated with an elastic response in this lake, with its modern diatom community returning to the one present before the 1970s. To evaluate this, we used a novel approach in which we integrated limnology, paleolimnology, and remote-sensing tools. Diatom assemblages were used to evaluate lake response to the external forcing represented by land use change and erosion, while remote images and magnetic susceptibility were used as a tool for quantifying these external forcings, therefore bridging the gap between aquatic and terrestrial dynamics.

## Site description

The STUX is a volcanic field in the Neotropics of eastern Mexico; it is an area of great ecological relevance because of its high biodiversity (Von Thaden et al. 2020), as it holds one of the last remnants of tropical evergreen rainforest in Mexico (Guevara and Laborde 2012). The STUX has two main volcanoes (San Martín Tuxtla and Santa Marta) and at least 368 monogenetic vents (Sieron



**Fig. 1** Location of the Sierra de Los Tuxtlas, eastern Mexico (inset) and of Lago Verde within the Sierra de Los Tuxtlas, Mexico. Red dots represent monogenetic volcanic vents (Sieron et al. 2021) and

blue dots represent maar lakes. Lago Verde is labeled and identified with an arrow. The dotted line delimits the Los Tuxtlas Biosphere Reserve

et al. 2023, 2021). Most of these monogenetic vents are cinder cones, but they also include maar craters, of which at least 13 contain a lake. Lago Verde ( $18^{\circ} 36' 43''$  N,  $95^{\circ} 20' 52''$  W, 149 m asl) is a maar lake on the western flanks of the San Martín Tuxtla volcano, on the STUX, outside of the limits of the Los Tuxtlas Biosphere Reserve (Fig. 1). The region has a mean annual temperature of  $25^{\circ}\text{C}$  and a mean annual precipitation of  $2500\text{ mm year}^{-1}$ . The lake is nearly circular, approximately 380 m in diameter, and has a maximum depth of 4 m (Vázquez et al. 2004). It lies in a small basin ( $0.45\text{ km}^2$ ), with steep rims on the north and western shores, and an elongated southeastern catchment drained by a small rivulet. In 2000, the lake was hypertrophic, characterized by turbid waters (Secchi disk depth  $= 0.4 \pm 0.2\text{ m}$ ) and a high chlorophyll *a* concentration ( $153 \pm 55\text{ mg m}^{-3}$ ). Grasslands for cattle ranching dominate the vegetation cover in this small basin, with conserved patches of secondary tropical trees (mainly *Bursera simaruba*, *Cecropia obtusifolia*, *Myriocarpa longipes*, and *Pachira aquatica*).

## Methods

Our methodological approach was to evaluate deforestation around the lake by analyzing remote images from different time windows from 1976 to 2024, and to contrast this information with limnological and paleolimnological data. Magnetic susceptibility was used as a physical proxy to evaluate sediment delivery to the lake in both modern (2023–2024) and historical (last ~200 years) sediment samples. The current lake conditions and diatom assemblages were studied during two annual cycles in 2023 and 2024, and contrasted with equivalent measurements taken in 2000 and 2003 (Vázquez and Caballero 2013; Vázquez et al. 2004), as well as with diatom assemblages from sediments dated to the last ~200 years (Caballero et al. 2006). Changes in diatom assemblages were used as indicators of lake response to external forcing; in this case, sediment delivery that changed lake transparency and nutrient loadings.

## Forest cover analysis

Remote images were analyzed to assess deforestation around Lago Verde, which has been the main driver of environmental change for this lake. Changes in forest cover over the last 50 years were analyzed using aerial photographs from 1976 and 2000 (INEGI 2024; <https://www.inegi.org.mx/app/geo2/siiv/>) and a high-resolution satellite image from 2024 (Planet Team 2022). It was not possible to locate aerial photographs dating to an earlier period. Land cover was classified into three major categories: forest, non-forest, and water bodies. All imagery was processed using the maximum likelihood algorithm in ArcGIS Pro 3.5.4 (ESRI 2025, <https://www.esri.com>). The classification results were compared to quantify changes in forest cover over time and to identify transition patterns between categories.

## Limnology

The lake was sampled during two annual cycles (May, September, and January) from May 2023 to January 2025. The samples collected in May 2023, September 2023, and January 2024 will be referred to as the 2023 annual cycle, while those collected in May 2024, September 2024, and January 2025 correspond to the 2024 annual cycle. The selected sampling months align with the strategy followed in the previous 2000 annual cycle, and they characterize the seasonal variation in the region; May represents the warmest season, September the wettest, and January the coldest. In situ limnological parameters were measured at two locations near the center of the lake, which were the same as those sampled in the 2000 study. Samples were collected from the surface (0.5 m) and bottom (3–3.5 m) using a Van Dorn bottle. Measurements of some variables were taken in situ on the surface and bottom using a multiparametric probe (Hydrolab DS5). These measurements included water temperature, electric conductivity (EC), and dissolved oxygen. Secchi disk visibility ( $Z_{SD}$ ) was also determined, and surface samples for water quality analyses were collected on each occasion. Water quality analysis included major ion determinations (carbonates, chlorides, sulphates,  $\text{Na}^+$ ,  $\text{K}^+$ ,  $\text{Ca}^{2+}$ , and  $\text{Mg}^{2+}$ ), chlorophyll *a* (Chla), total phosphorus (TP), soluble reactive phosphorus (SRP), ammonium, nitrite, and nitrate. Total dissolved nitrogen (DIN) was estimated by adding ammonium, nitrite, and nitrate determinations. The samples were refrigerated until analyzed using standard methodologies (American Public Health Association (APHA) et al. 2005). Samples for cation determinations were acidified with  $\text{HNO}_3$ . Samples for Chla determinations were filtered (Whatman GF/C filters), Chla was extracted with 90% methanol, and the extracts were measured spectrophotometrically; concentrations were expressed as  $\text{mg m}^{-3}$ . Samples for ammonium and nitrate were acidified with  $\text{H}_2\text{SO}_4$ . Nutrients

were determined in a Thermo Scientific GENESYS 20 visible spectrophotometer (ammonium by Nessler's method, nitrite by diazotization, nitrate by the brucine colorimetric method, TP by persulfate digestion, SRP by the ascorbic acid method) (APHA et al. 2005).

The trophic level index (TLI) based on Chla and DIN concentrations was used to evaluate the changes in trophic conditions of Lago Verde. This index was selected because unlike other indexes that are based on Carlson's Trophic state index (Carlson 1977), this one was developed for lakes with high phosphorous concentrations, where nitrogen becomes the limiting element, as in Lago Verde (Kratzer and Brezonik 1981; Paulic et al. 1996). The TLI was estimated for the 2023–2024 annual cycle samples, and also for the previously published data collected in 2000 (January, May, and September) and July 2003 (Vázquez and Caballero 2013; Vázquez et al. 2004). The LSI limits used were: mesotrophic ( $40 < \text{TLI} \leq 50$ ), eutrophic ( $50 < \text{TLI} \leq 70$ ), and hypereutrophic ( $\text{TLI} > 70$ ), and they were calculated using the following formulas (Paulic et al. 1996):

$$\text{TLI}_{(\text{Chla})} = 16.8 + [14.4 * \ln(\text{Chla})] \quad (1)$$

$$\text{TLI}_{(\text{DIN})} = 10 * [5.96 + 2.15 * \ln(\text{DIN} + 0.001)] \quad (2)$$

$$\text{TLI} = (\text{TSI}_{(\text{Chla})} + \text{TSI}_{(\text{DIN})}) / 2 \quad (3)$$

## Diatom analyses

Diatoms are widely used as bioindicators in modern and paleolimnological studies because they meet the following criteria: (i) their populations change rapidly in response to environmental changes due to their short life cycles, (ii) their species have well-documented ecological niches, and (iii) their siliceous remains (valves) allow for an accurate taxonomic identification (Battarbee et al. 2001; Caballero and Maidana 2017; Smol and Cumming 2000). Furthermore, after dying, the siliceous diatom valves are deposited in the sediments where they are preserved; therefore they are among the few biological groups that can be sampled and analyzed identically in both modern and old (archive) sediments. In the context of this work, this is particularly relevant as the past diatom assemblages provided the necessary historical context for interpretation of the modern assemblages collected in water and sediment samples.

## Diatoms in water samples

Surface (0.5 m) water samples for quantitative diatom analysis were collected from two locations near the center of the lake using 1 L plastic bottles, on the same dates and at the

same sites as the limnological parameters and water quality samples. They were fixed with 1% Lugol's iodine solution and quantified using the Utermöhl method (Hasle 1978) with an inverted microscope (Leica LDMS). A fraction of these samples was oxidized following the same method as for surface sediment samples to allow for more detailed taxonomic observations and taxonomic harmonization with the sediment samples. Diatoms were identified to species level using specialized literature (Gasse 1986; Krammer and Lange-Bertalot, 1986).

### Diatoms in surface sediments

Surface sediment samples for diatom analysis and magnetic susceptibility measurements were collected from the central part of the lake using an Ekman dredge on the same dates as the previous samples. Sediment samples for diatom analysis were dried, and 0.2 g of dry sediment was treated with HCl (10%) and H<sub>2</sub>O<sub>2</sub> (30%) to eliminate carbonates and organic matter (Battarbee 1986; Schrader 1973). Slides were prepared with 200 µl of the clean suspension and mounted with Naphrax. Diatom counts of at least 400 valves were undertaken using an Olympus BX50 microscope at 1000× magnification. Diatom species were identified using specialized literature (Krammer and Lange-Bertalot, 1986; Spaulding et al. 2021). Results were expressed as species relative abundances (%).

### Magnetic susceptibility of surface sediment samples

Magnetic susceptibility was measured in the same sediment samples collected in September 2023 and September 2024 for diatom analysis, which allowed for a direct correlation between physical (magnetic susceptibility) and biological (diatoms) variables. Magnetic susceptibility is a physical parameter that responds to the entrance of allochthonous volcanic sediment from the surrounding slopes, and can be used as a proxy for sediment delivery to the lake (Michaud et al. 2006; Thompson et al. 1975). The same methodology was used for the core sediment samples in our previous work (Ortega-Guerrero et al. 2006). For this purpose, 2 cm side cubes were filled with sediment, and their net weight (A&D HF2000G balance) and bulk density (g cm<sup>-3</sup>) were determined. Magnetic susceptibility was measured using a Bartington MS2B system, and the mass-normalized magnetic susceptibility ( $\chi$ ) was determined by dividing by the sediment density; the results were expressed as 10<sup>-6</sup> m<sup>3</sup> kg<sup>-1</sup>.

### Statistical analysis

Annual averages and standard deviations of EC, DIN, TP, Z<sub>SD</sub>, Chla, and LTSI were estimated. Their values were compared with those previously reported for the 2000 annual

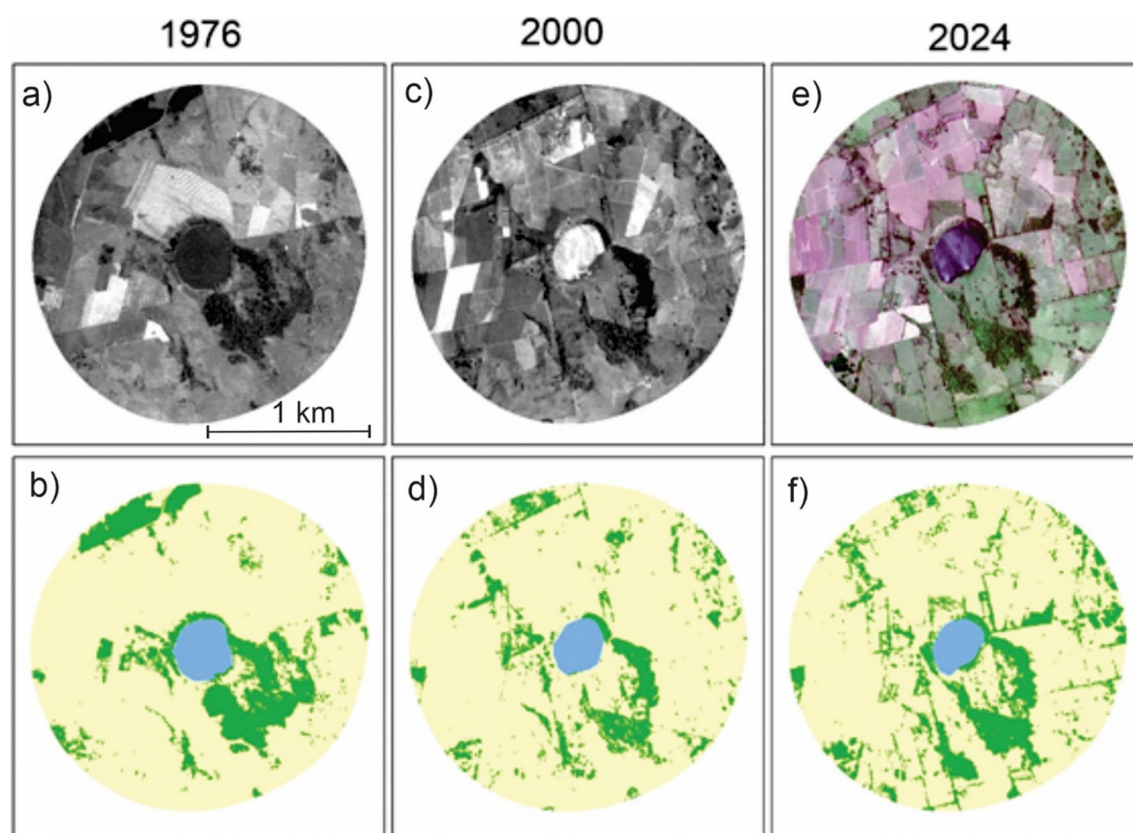
cycle, when samples were collected on similar dates (January, May, and September) and with the single measurements made in July 2003, using a one-way analysis of variance (ANOVA). This test was performed in paleontological statistics (PAST) software (Hammer et al. 2001).

To identify trends in the diatom assemblages in the water samples and the environmental variables (EC, temperature, TP, Chla, DIN, pH, and Z<sub>SD</sub>), a canonical correspondence analysis (CCA) was performed (ter Braak 1986), and a series of partial CCAs were also run using single environmental variables to evaluate their importance in explaining samples and species distribution. The long-term trends in the diatom assemblages in sediment samples from Lago Verde were visualized using a detrended correspondence analyses (DCA) to measure the distance of the 2023 and 2024 samples to those previously studied (Correa-Metrio et al. 2014). These analyses were run with the Vegan package in R (R Development Core Team 2020) using the square root transformation of the environmental variables and diatom densities (water samples), and species relative abundances (sediment samples). For the water column samples, analysis included the six samples corresponding to the 2023 and 2024 annual cycles, and the previously published samples collected in July 2003 (Vázquez and Caballero 2013) and in 2000 (January, May, and September), providing a 25-year long perspective. For the sediment samples, analysis included the six samples corresponding to the 2023 and 2024 annual cycles, the previously published sample collected in July 2003 (Vázquez and Caballero 2013), and the top 60 cm (10 samples) of the <sup>210</sup>Pb-dated core sequence collected in 2001, with estimated ages from ~1808 to 2001 (Caballero et al. 2006), providing an ~200-year perspective of lake evolution. The scores of a DCA ordination represent standard deviations of ecological turnover (Correa-Metrio et al. 2014; Gauch 1982). Therefore, the distance between samples in the ecological space defined by the first two axes of the DCA gives an accurate perspective of the species turnover between samples, considering that ~50% ecological turnover is represented by each unit. However, to obtain a more accurate estimate of species turnover between samples, the Bray–Curtis dissimilarity was also estimated for water and sediment samples. This metric varies between 0 (no species turnover) and 1 (100% or total species turnover).

## Results

### Forest cover

Forest cover in the 1 km buffer zone around Lago Verde was already low in 1976 (Fig. 2a, b), occupying ~17% of the area (73.5 ha), while non-forested land dominated the landscape (Table 1). Forest cover continued to decline to



**Fig. 2** Remote images with a 1 km buffer zone, dating to 1976 (**a** and **b**), 2000 (**c** and **d**), and 2024 (**e** and **f**) from Lago Verde, Sierra de Los Tuxtlas, Mexico. In (**b**), (**d**), and (**f**), green shading represents

forested areas, yellow shading corresponds with non-forested areas, and blue shading represents the lake surface

**Table 1** Surface area (ha) corresponding to forested, non-forested, and lake areas, as well as the percentage of forested areas from remote images dating to 1976, 2000, and 2024 from Lago Verde, Sierra de Los Tuxtlas, Mexico

| Year | Forested (ha) | Non-forested (ha) | Lake (ha) | Forested (%) |
|------|---------------|-------------------|-----------|--------------|
| 1976 | 73.5          | 354.1             | 14.8      | 17           |
| 2000 | 55.0          | 376.2             | 11.2      | 12           |
| 2024 | 80.3          | 351.5             | 10.7      | 18           |

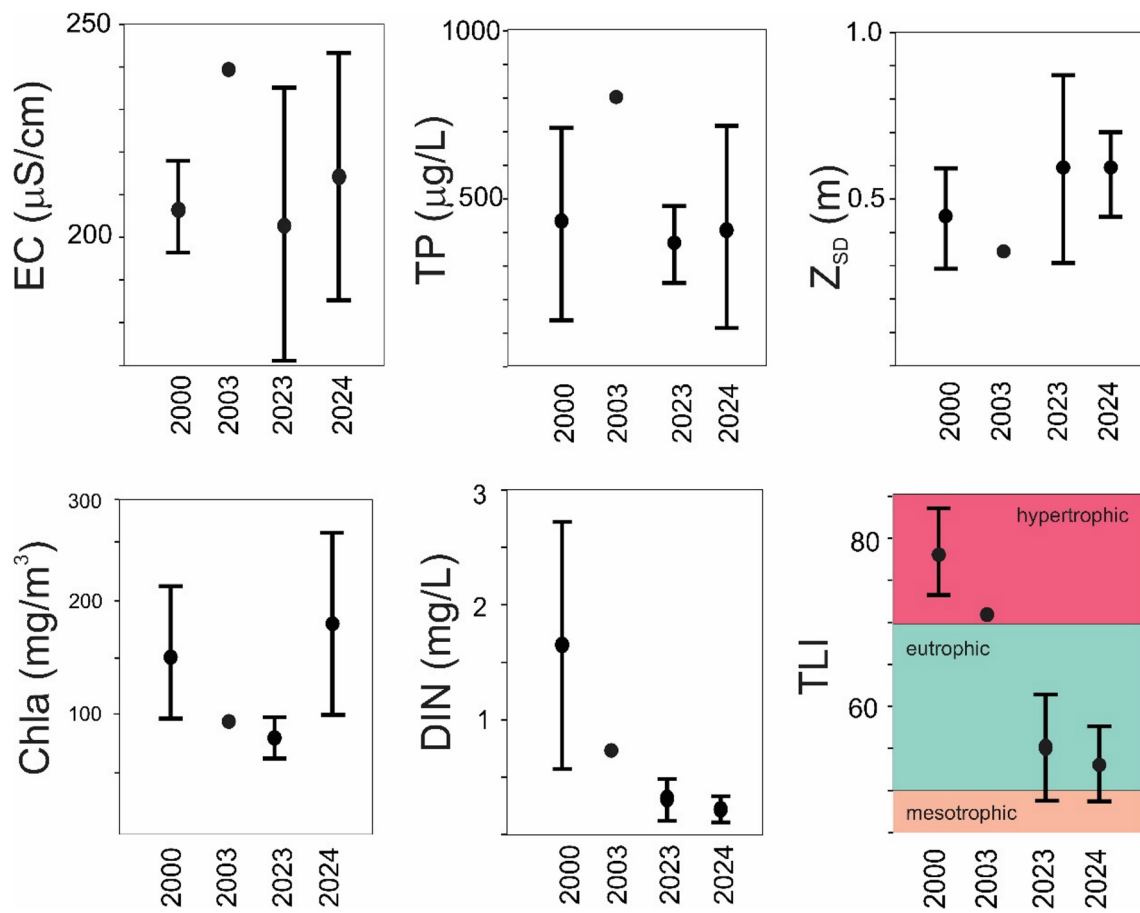
12% (55 ha) by 2000 (Fig. 2c, d), indicating continued land-use changes during the late twentieth century. Nevertheless, in 2024 (Fig. 2e, f) forest cover increased slightly to 18% (80 ha), indicating partial vegetation recovery, while non-forested areas reduced from 84.2% in 2000 to 79.3%. These shifts may reflect recent declines in agricultural and grazing activities and a gradual expansion of vegetation over abandoned lands. Further, the lake surface decreased from ~15 ha in 1976 to ~11 ha in 2000. This reduction affected the southeastern sector of Lago Verde, where the small stream flows into the lake. The sediment delivery from this small

inflowing stream has progressively clogged this part of the lake, contributing to its present configuration.

Remote images of this lake analyzed by Caballero et al. (2006) identified a reduction in forest cover occurring between 1980 and 1991. In these images, only the area corresponding to the lake's hydrological basin was analyzed; however, the 1976 image shows the same forest cover within the lake basin as in 1980, while the 2000 image has the same forest cover within the lake basin as in 1991 (Electronic Supplementary Material (ESM) Fig. 1). The new remote images analyzed in this work allowed for a more detailed timeline to be established of changes in the extent of the surrounding vegetation over the past five decades, demonstrating high deforestation since the 1970s as well as a progressive clogging of the eastern part of the lake that reduced its surface during the 1990s.

### Trophic level parameters

The average values and standard deviations for the 2023 and 2024 annual cycle measurements for EC, DIN, TP,  $Z_{SD}$ , Chla, and LTSI are presented in Fig. 3, where they are



**Fig. 3** Average values and standard deviations for electric conductivity (EC), total phosphorous (TP), Secchi disk visibility ( $Z_{SD}$ ), chlorophyll *a* (Chla), dissolved inorganic nitrogen (DIN), and trophic level index (TLI) for the 2000 (January, May, September), 2023, and 2024 (May, September, January) annual cycles, as well as for July 2003,

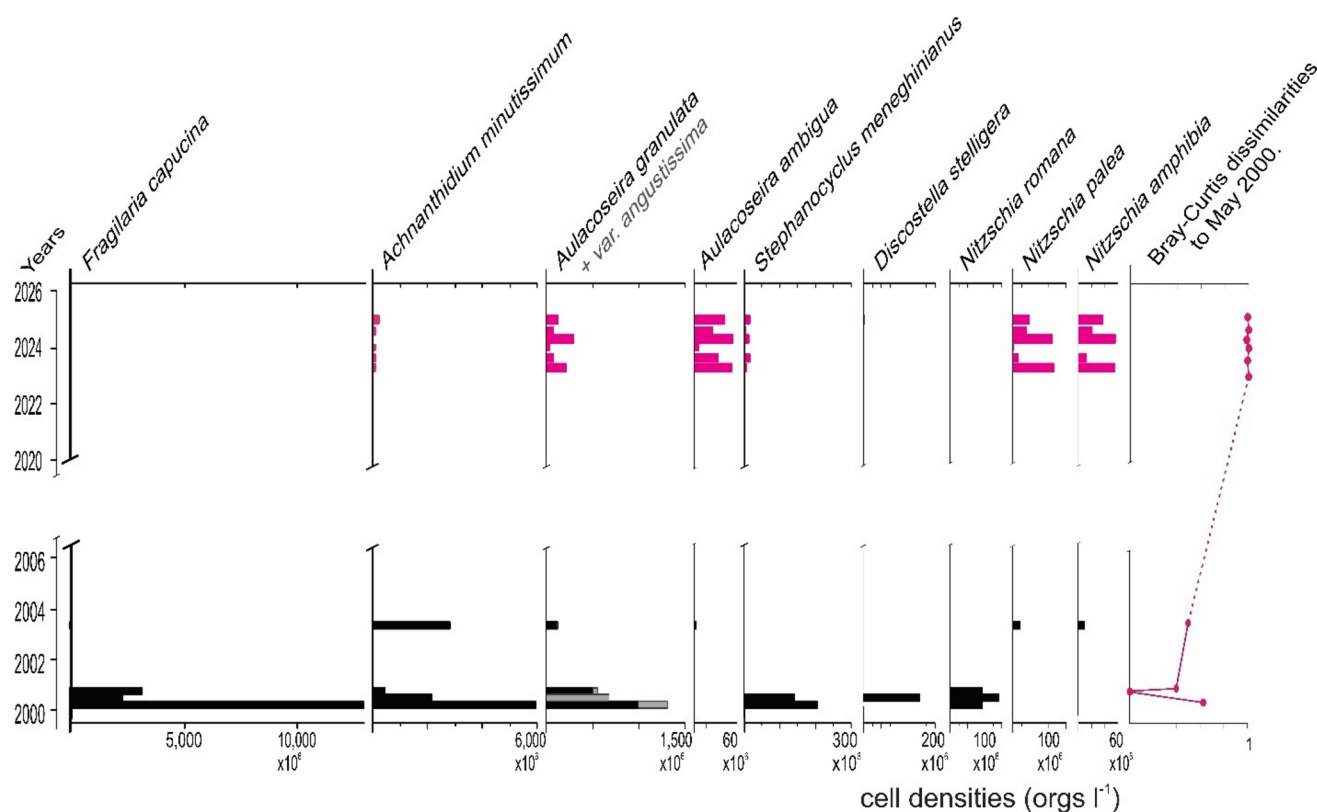
from Lago Verde, Sierra de Los Tuxtlas, Mexico. Data from 2000 and 2003 were obtained from Vázquez et al. (2004) and Vázquez and Caballero (2013), respectively. ANOVA tests indicated that only DIN and TLI values were statistically different between the years analyzed

contrasted against the 2000 and 2003 values. The ANOVA tests showed that there were no significant differences in EC, TP,  $Z_{SD}$ , and Chla between the 2000 and the 2023 and 2024 annual cycles ( $p > 0.08$ ), but for DIN and TLI there were differences between the measurements ( $p \leq 0.05$ ). DIN showed lower values in 2023 and 2024 (average  $0.3 \pm 0.2$  and  $0.2 \pm 0.1 \text{ mg L}^{-1}$  respectively) than in 2000 (average  $1.6 \pm 1 \text{ mg L}^{-1}$ , Fig. 3). Furthermore, the TSI showed that Lago Verde was in the hypereutrophic range in 2000 and 2003, but that it reduced to eutrophic in 2023 and 2024. The data indicated that there was no significant change during the last 25 years in lake transparency ( $Z_{SD}$ ), TP, or Chla, but that there was a significant reduction in nitrogen concentrations from annual average values that exceeded  $1.5 \text{ mg L}^{-1}$  in 2000 to  $0.2 \text{ mg L}^{-1}$  in 2024. This can be also visualized as a change from hypertrophic to eutrophic conditions according to the Chla and DIN based TLI.

### Diatoms in water samples

A total of 26 diatom taxa were identified in the water samples collected from Lago Verde during the 2000, 2023, and 2024 annual cycles (ESM Table 1). The higher cell densities were concentrated in ten taxa (Fig. 4): *Fragilaria capucina*, *Achnantheidium minutissimum*, *Aulacoseira granulata*, *A. granulata* var. *angustissima*, *A. ambigua*, *Stephanocyclus meneghinianus*, *Discostella stelligera*, *Nitzschia romana*, *N. palea*, and *N. amphibia*.

In the CCA analysis performed on these samples, axis 1 accounted for 53% of the variance, and axis 2 for 20%. Axis 1 is negatively correlated with DIN while axis 2 seems to be a combination of several environmental variables; it is negatively correlated with Chla and positively with  $Z_{SD}$ . The partial CCAs showed, however, that only DIN was significant in explaining diatom species and sample distribution ( $p = 0.02$ ). The samples collected during the 2000 annual cycle



**Fig. 4** Diatom species cell densities and Bray–Curtis dissimilarities in the water samples from Lago Verde, Sierra de Los Tuxtlas, Mexico, during the 2000, 2023, and 2024 annual cycles, as well as in July 2003. Note the changes in scales between species. Black bars corre-

spond to previously published data (Vázquez et al. 2004 and Vázquez and Caballero 2013), while pink bars correspond to the new data presented in this paper

had negative axis 1 scores and correlated with the highest DIN values (Fig. 5a). These samples were associated with the highest densities of *Fragilaria capucina*, as well as with high densities of *Achnanthyidum minutissimum*, *Aulacoseira granulata* var. *angustissima*, and *Discostella stelligera* (Figs. 4 and 5B). The sample from May 2000 separated from the January and September samples, with negative axis 2 scores, following higher Chla values. On the contrary, the samples collected during the 2023 and 2024 annual cycles had high positive axis 1 scores and were located at the lower end of the DIN gradient. These samples were associated with high densities of *Aulacoseira ambigua* and *A. granulata*, as well as of *Nitzschia amphibia* and *N. palea*, species that also had high axis 1 scores. The sample from July 2003 was located at an intermediate position between these two groups, and was characterized by the highest density of *Achnanthyidum minutissimum* in all the samples.

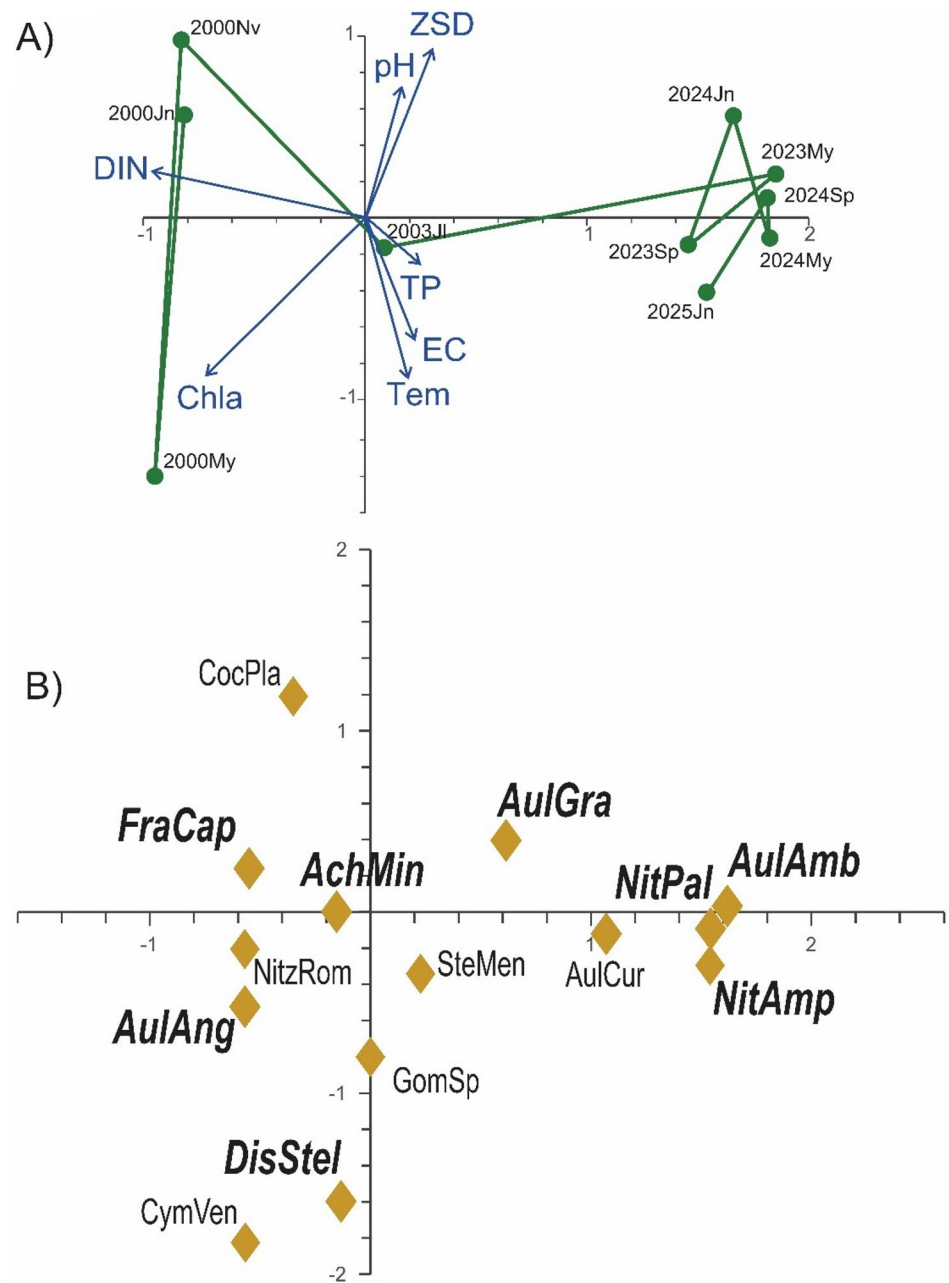
The Bray–Curtis dissimilarities (ESM Table 2) between the 2000 and 2023–2024 annual cycles reflect a total species replacement (distances range from 0.9 to 1, Fig. 4), showing a shift in lake conditions during the last 25 years. In the ecological space defined by the first two CCA axes, this trend is also evident (Fig. 5), and is associated with a reduction

in the cell densities of *Fragilaria capucina* and *A. minutissimum*, which had their highest concentrations in 2000 but nearly disappeared in the samples from the 2023 and 2024 annual cycles. In contrast, in the 2023 and 2024 samples there was an increase in *Aulacoseira ambigua*, *A. granulata*, *Nitzschia palea*, and *N. amphibia* (Figs. 4, 5). These changes follow the reduced DIN concentrations observed between the 2000 and the 2023–2024 samples.

### Diatoms and magnetic susceptibility in sediment samples

A total of 22 diatom taxa were identified in the sediment samples collected during the 2023 and 2024 annual cycles in Lago Verde (ESM Table 3). However, only seven had abundances higher than 5% in at least one sample (Fig. 6); the most abundant were *Aulacoseira granulata*, *A. ambigua*, *Fragilaria capucina*, and *Achnanthyidum minutissimum*. These four taxa were also the most abundant in the water samples from 2023 and 2024, showing consistency between both kinds of samples. Magnetic susceptibility ( $\chi$ ) of the 2023 and 2024 samples was lower than  $1 \times 10^{-6} \text{ m}^3 \text{ kg}^{-1}$ ,

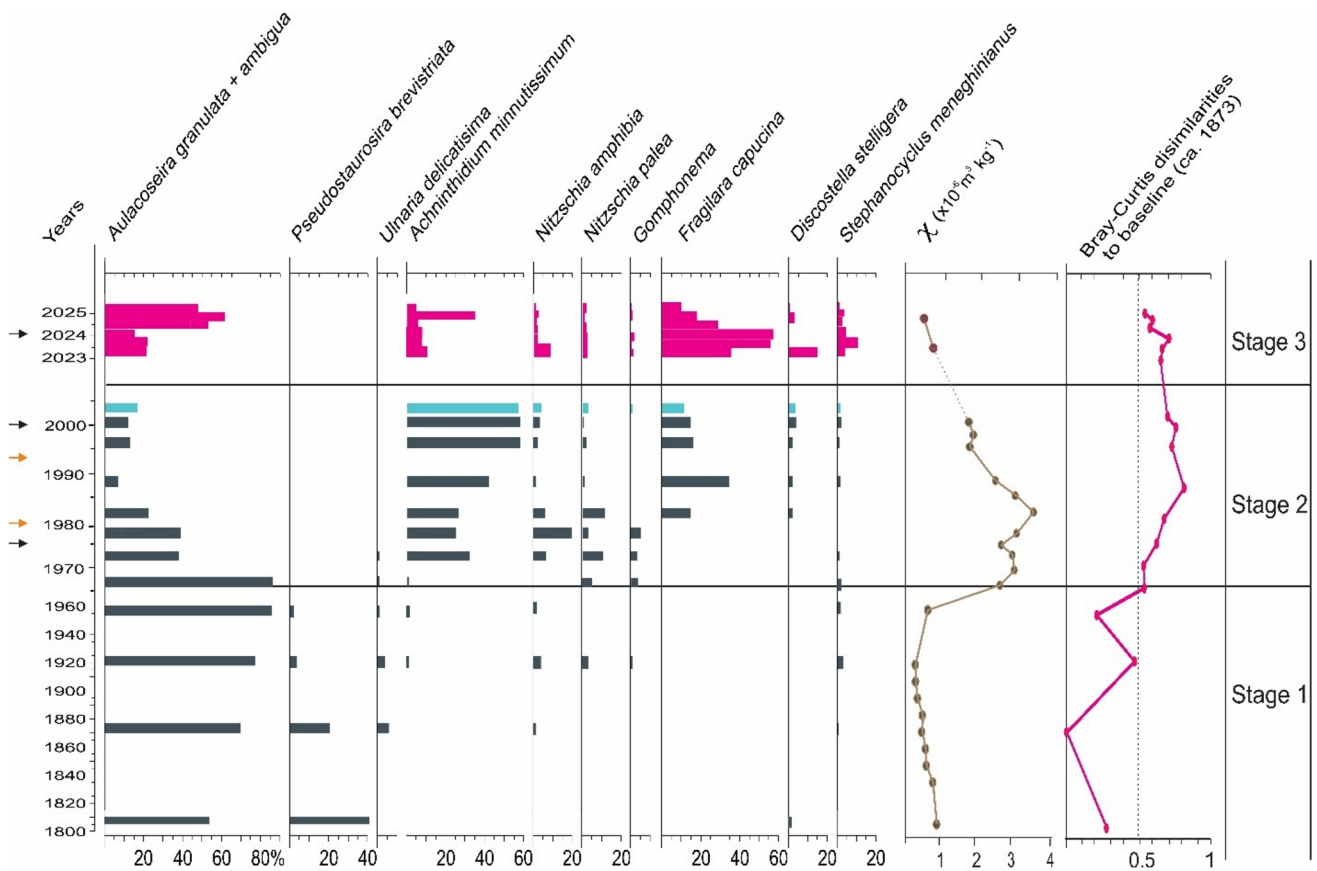
**Fig. 5** Plot of the CCA axis 1 versus axis 2 for the water-column environmental variables and diatom assemblages of Lago Verde, Sierra de Los Tuxtlas, Mexico, from 2000, 2023, and 2024 annual cycles as well as from July 2003. (a) Sample scores and environmental variable vectors, (b) species scores; DIN dissolved inorganic nitrogen,  $Z_{SD}$  Secchi disk visibility, TP total phosphorous, EC electric conductivity, Tem temperature, Chla chlorophyll *a*, *Jn* January, *My* May, *Jl* July, *Sp* September, *Nv* November. Species abbreviations listed in ESM Table 1



which contrasts with the higher values in the sediments dating between 1968 and 1988, and even 2001 (Fig. 6).

*Aulacoseira granulata* and *A. ambigua* were the most abundant taxa in the sediment samples in Lago Verde before peak disturbance in the 1970s (Fig. 6); these species are cosmopolitan, planktonic diatoms, common in central and southern Mexico, in freshwater lakes ranging from mesotrophic to eutrophic (Avendaño et al. 2023; Caballero et al. 2025; Ehrlich 1995; Krammer and Lange-Bertalot 1991; Ramírez-Nava et al. 2022; Stoermer et al. 1996). These species are considered to indicate a lower trophic level than those found during the earlier period of peak lake disturbance in the 1970s and 1980s. On the contrary,

*Achnantheidium minutissimum* is characteristic of the sediment samples during that peak disturbance. This is a cosmopolitan, periphytic, or tychoplanktonic taxon that is particularly abundant in the disturbed eutrophic to hypertrophic maar lakes at the STUX (Vázquez and Caballero 2013). It is considered by some authors as a primary colonizer in disturbed environments (Hodgson et al. 1997; Peterson and Stevenson 1992), and in other lakes in central Mexico it has also been associated with human impact and higher erosion rates (Rodríguez-Ramírez et al. 2015). Its distribution could be favored by high nutrient levels, particularly nitrogen, as indicated by our field data from 2000 and 2003 and its position on the CCA diagram (Fig. 5). *F. capucina* is also an



**Fig. 6** Relative abundances of the main diatom species, magnetic susceptibility ( $\chi$ ), and Bray–Curtis dissimilarities to baseline (~1873) in sedimentary samples dating from ~1808 to January 2025 from Lago Verde, Sierra de Los Tuxtlas, Mexico. Data from samples dating from 1808 and 2001 (black bars) were published in Caballero et al. (2006), data from 2003 (green bars) were published in Vázquez and Caballero

(2013), and the top samples (pink bars) were collected between May 2023 and January 2025 (this paper). Black arrows on the left correspond to the years of the remote images that were analyzed in this paper. Golden arrows on the left correspond to remote images analyzed in Caballero et al. (2006)

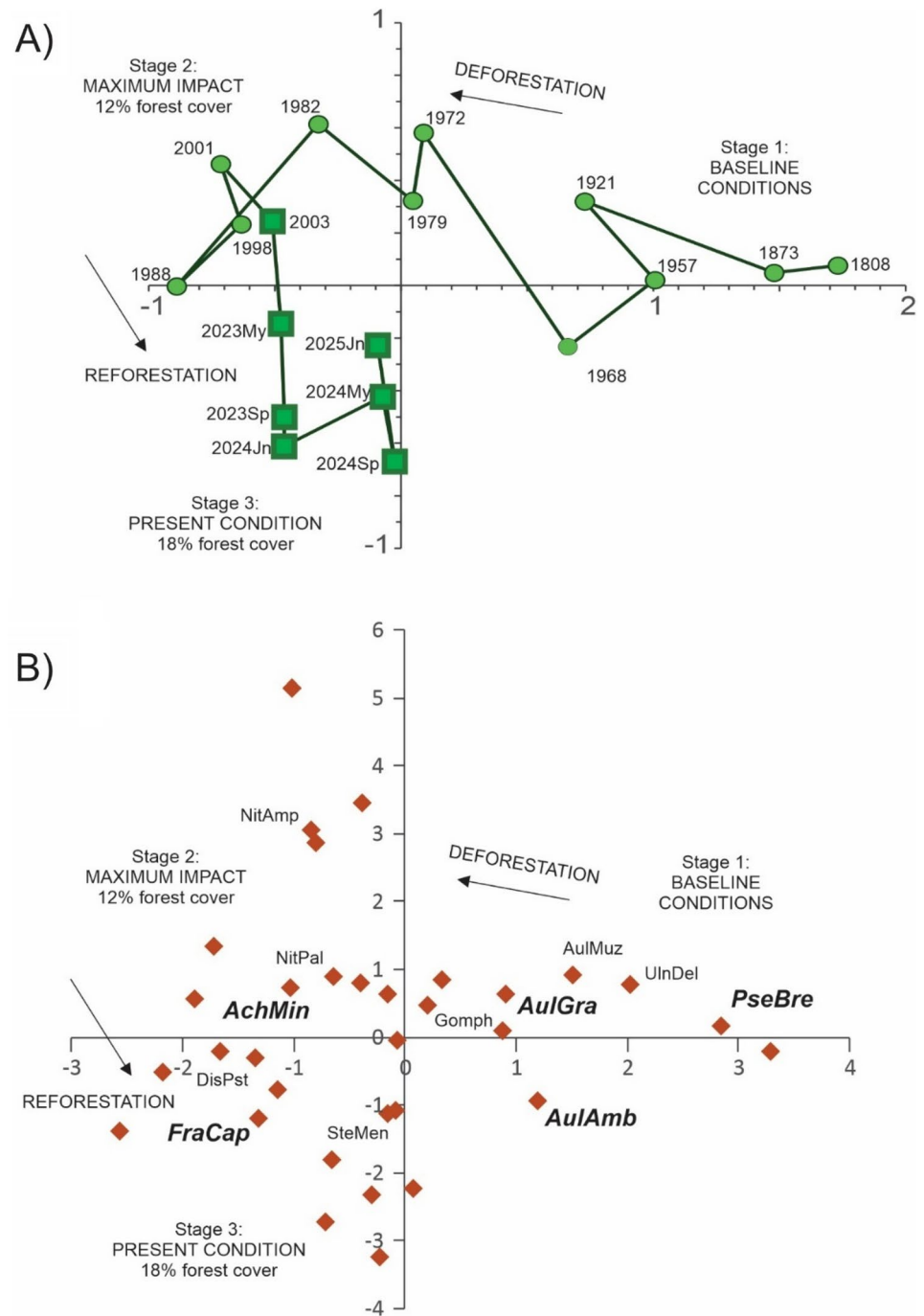
important taxon in Lago Verde, which became abundant after the peak disturbance in the 1980s. This is a cosmopolitan, periphytic, tycho planktonic, or planktonic taxon, tolerant of high nutrient levels (Bennion 1995; Ehrlich 1995; Gasse 1986; Hall et al. 1999; Stoermer et al. 1996), but unlike the previous taxa, this species is uncommon in other lakes in central and southern Mexico, including those at the STUX (Vázquez and Caballero 2013). According to the CCA on the water diatom samples, this species is associated with higher DIN values.

The DCA analysis performed on the sediment samples from Lago Verde dating to the last ~200 years showed long-term trends in the diatom assemblages (Fig. 7). Axis 1 explained about 36% of the total variance, while axis 2 explained about 15%. The DCA axis 1 versus axis 2 ecological space showed a separation of the sediment samples dating to before the 1970s, as they formed a group on the high (>0.6) positive axis 1 values (Fig. 7a). These samples represent the reference or baseline condition of the lake prior

to modern deforestation (stage 1), with a diatom assemblage composed mainly of *Aulacoseira granulata*, *A. ambigua*, and *Pseudostaurosira brevistriata* (Figs. 6 and 7b). The two samples dating to the 1970s were located at lower positive values on axis 1 (<0.3), while the group of samples dating from the 1980s to 2003 formed a cluster with negative axis 1 scores and positive axis 2 scores (stage 2).

The 1970s–2003 (stage 2) samples were characterized by the highest magnetic susceptibility values ( $> 1.5 \times 10^{-6} \text{ m}^3 \text{ kg}^{-1}$ ) (Fig. 6), which allowed us to infer that erosion rates impacting the lake were highest during this time (Caballero et al. 2006). These samples had low abundance of *Aulacoseira* spp. and the highest percentages of *Achnanthes minutissimum* (Figs. 6 and 7b). This last species is representative of an impacted state of the lake (stage 2). Magnetic susceptibility values declined during the 1990s–2003, when *F. capucina* also became part of the diatom assemblage. Magnetic susceptibility returned to lower values in 2023 and 2024, and in the ecological space defined by DCA axis

**Fig. 7** Plot of the DCA axis 1 versus axis 2 for the diatom assemblages preserved in sediment samples dating from ~1808 to 2025 from Lago Verde, Sierra de Los Tuxtlas, Mexico. **(a)** Sample scores; circles are core sediment samples, squares are dredge sediment samples. **(b)** Species scores, bold text corresponds with dominant taxa. Species abbreviations are listed in ESM Table 1



1 versus axis 2, these samples (stage 3) formed a separate group on the negative side of both axes—with an assemblage that contrasts with the previous group of samples by lower abundances of *A. minutissimum*, high-to-medium percentages of *F. capucina*, and higher percentages of *Aulacoseira* spp., particularly in the samples from the 2024 annual cycle (Figs. 6 and 7B).

The Bray–Curtis dissimilarities (ESM Table 4) between the baseline samples of stage 1 showed medium-to-low (<0.5) species turnover (Fig. 6). Their distribution on

the DCA axis 1 versus axis 2 ecological space showed a gradual transition toward the left of the diagram, which represents the most impacted conditions (Figs. 6, 7). The exception was the ~1957 sample, which presented an elastic response, returning to higher (baseline) axis 1 scores and showing lower Bray–Curtis dissimilarity. This shows that the lake's baseline condition was not stable (low resistance), as it gradually changed, with linear and elastic transitions. The Bray–Curtis distance between the baseline conditions of stage 1 and the maximum impact during stage 2, in the

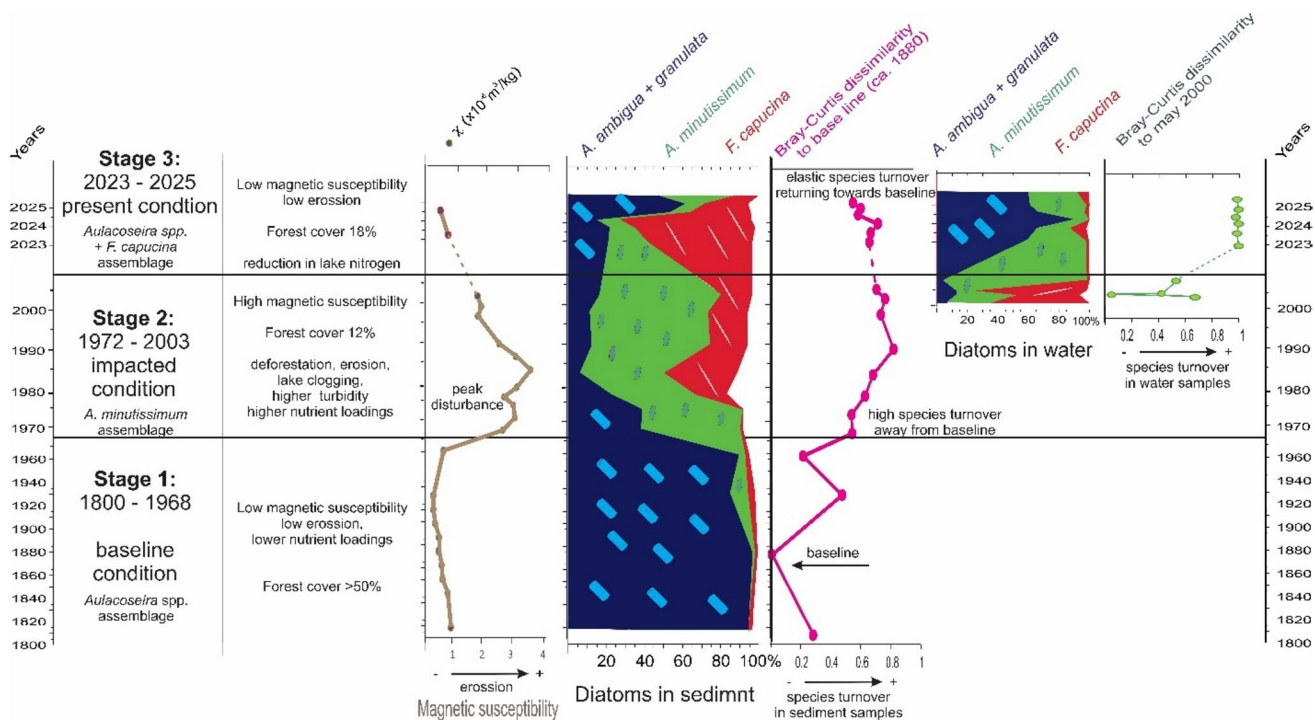
1980s, was high ( $> 0.5$ – $0.8$ , Fig. 6), showing a high species turnover and reflecting a rapid shift in lake condition. The larger distance between samples in the DCA ecological space indicates that this shift occurred between ~1968 and ~1972, as magnetic susceptibility increased, showing the onset of peak disturbance by deforestation around Lago Verde (stage 2). The most recent samples from this lake, collected during a time of landscape stabilization and reforestation, showed lower Bray–Curtis dissimilarities to baseline conditions (0.5 to 0.6). Furthermore, these recent samples are in a different region of the DCA ecological space (stage 3), with a trend toward higher values along axis 1 (toward baseline). This indicates that in the last 25 years the lake has changed, with linear to elastic responses, toward a new condition where *Aulacoseira granulata* and *A. ambigua* have recovered, but where *F. capucina* and low numbers of *A. minutissimum* are also present. This new condition is, however, closer to its pre-disturbance baseline.

## Discussion

Lago Verde, in the STUX region of eastern Mexico, is an ideal site for studying aquatic ecosystem changes through time associated with forest clearance disturbance in the Neotropics. The sediments from Lago Verde provide an excellent record of land use changes (deforestation) through the magnetic susceptibility signal, while diatom associations

reflect the lake's internal dynamics and show state shifts occurring in the lake during the last ~200 years. This long-term approach is further supported in this work by remote images covering the last ~50 years, and field data recording the lake's trophic status and the diatom assemblages in water samples during the last 25 years (Fig. 8).

Remote images showed that by 1976, there was already an ongoing deforestation process that intensified during the 1980s, in agreement with historical information on high deforestation rates in the STUX during the 1970s and 1980s (Guevara et al. 2004). The magnetic susceptibility record of Lago Verde showed high erosion rates affecting this lake between ~1968 and ~1988 (Ortega-Guerrero et al. 2006), identifying the time of maximum disturbance. This disturbance was also related to progressive clogging of the eastern part of the lake that receives discharge from a stream, leading to a reduction in the lake's surface during the 1990s. After peak disturbance, the magnetic susceptibility decreased while forest cover around the lake remained relatively stable during the late 1990s, with reforestation occurring during the last two decades. The pattern of accelerated deforestation during the 1970s with a later slowdown during the 2000s is common to other important Neotropical regions such as the Amazon Basin (Rosa et al. 2016). However, despite the recovery in the forest cover around the lake, field measurements showed that the lake remained turbid, albeit with significantly lower nitrogen levels. This reduction in DIN was the main environmental factor associated



**Fig. 8** Summary figure of the three stages of Lago Verde identified during the last 200 years

with the observed changes in diatom species over the last 25 years as the lake changed from hypertrophic to eutrophic. Furthermore, nitrogen concentrations in Lago Verde during 2023 and 2024 were seasonally below the  $0.15 \text{ mg L}^{-1}$  starvation limit proposed by Reynolds (1999). Lower nitrogen levels can have cascading effects in the algal community of phosphorus rich lakes, favoring species that are more competitive under low nitrogen conditions (Hayes et al. 2015).

The slowdown in deforestation during the 1990s and the recent reforestation trend provided an excellent opportunity to explore the response of a Neotropical lake ecosystem before, during, and after disturbance. Our results show that Lago Verde experienced gradual to elastic responses during periods of low deforestation pressure in the late nineteenth and early twentieth centuries (stage 1, Fig. 8), suggesting relatively low resistance to disturbance. Furthermore, lake conditions shifted over a relatively short time interval (possible threshold transition) when disturbance peaked in the 1970s (stage 2, Fig. 8). This was the largest change in the lake documented in the last 200 years. However, it is difficult to establish whether this represented a threshold change since the time resolution between the analyzed core samples was around 4 years, even though this change could have happened in a much shorter time.

High species turnover was identified in the water samples diatom assemblages during the slowdown in deforestation occurring in the last 25 years (Fig. 8), with a nearly total reduction in the abundance of *F. capucina* and *A. minutissimum*. The diatom assemblages in sediment samples offer a more conservative perspective, with a partial turnover scenario during this same time, but with a similar pattern of decreasing values of *F. capucina* and *A. minutissima*, while *Aulacoseria* spp. increased. These differences between assemblage changes in the water and sediment samples can be explained because the sediments integrate information from a longer period (~6 months to 1 year), in contrast to the instantaneous information from the water samples. In both cases, however, the evidence points to an ongoing transition in lake conditions that involves a reduction in the disturbance taxa (e.g., *A. minutissimum*), while the baseline taxa (e.g., *Aulacoseira* spp.) are recovering (stage 3, Fig. 8). This is an ongoing elastic response (resilience), where the lake has reached a new condition (*Aulacoseria* spp. and *F. capucina*) that is closer to its baseline. According to this interpretation, our initial hypothesis was partially correct, as after 40 years of peak disturbance, the lake has reached a new state that is closer to its baseline condition (long-term resilience).

Long-term resilience processes such as the one observed in Lago Verde can be favored by geographic diversity, as species can disperse to adjacent regions where environmental conditions remain favorable (i.e., better climate or lower human impact) and then function as a dispersion source for

repopulation after disturbance, thereby acting as metacommunity units (Benito et al. 2018). In this way, the intricate topography of central Mexico represents a favorable element for ecosystem resilience.

The long-term resilience pattern of change in Lago Verde, with a partial return to baseline conditions, is comparable to what was observed in Lake Ontario (easternmost of the Laurentian Great Lakes), where peak disturbance during the early twentieth century transformed the lake from its baseline condition to a eutrophic state where pollution tolerant diatom taxa proliferated. Thanks to pollution abatement programs, the lake recovered during the 1980s, lowering its phosphorous levels with a recovery of baseline diatom taxa, but also with the incorporation of small centric species, which were absent before disturbance and which have been associated in the Laurentian Great Lakes with changing (warmer) climatic conditions (Pillsbury et al. 2021).

The Lago Verde paleolimnological sequence also recorded a previous disturbance event that offers the opportunity to compare the lake's recent response (last 200 years) to disturbance in the more distant past (last 2000 years), and to evaluate another interesting question in ecology: whether ecosystem changes associated with human impact during the last century are unprecedented (Benito et al. 2022). In the STUX, humans have been present for several millennia (Stoner 2017), and specifically around Lago Verde, the presence of *Zea mays* pollen attested for past agricultural activities (Lozano-García et al. 2010). The previous human disturbance event in Lago Verde occurred during the Classic period, between AD 100 and 800, and was also associated with deforestation, high erosion, and high magnetic susceptibility (Lozano-García et al. 2010). During the Classic period disruption event, the lake responded in a way that was very similar to what happened in the 1970s, with a rapid shift in lake conditions toward an impacted state that was dominated by *A. minutissimum*. In this case, the baseline condition of the lake was defined by the dominance of *Discostella stelligera*, a diatom species that disappeared from the record during its impacted interval (AD 100–800). However, in this case, the impacted condition persisted for nearly 600 years until agricultural activity, erosion, and magnetic susceptibility declined around AD 800, and as climate changed toward wetter conditions and human populations in the region declined. At that time, the lake experienced another rapid state shift when baseline *D. stelligera* recovered, but only in low numbers, and a new arrival, *Pseudostaurosira brevistriata*, became dominant. The impacted state of Lago Verde during the Classic period represents a clear precedent of human-induced disturbance that was not only comparable in intensity to modern processes, but that also lasted longer than the modern one (~600 years versus ~30 years). Furthermore, the partial recovery of baseline conditions after disturbance was similar to what happened in Lake Ontario,

and in both cases, it was associated with changing climatic conditions that precluded a full elastic response, as post-impact conditions are different from those at baseline. This is of relevance for understanding contemporary responses to human impact, as modern ecosystems are also under the stress of a changing, global warming scenario. In the case of Lago Verde, *F. capucina* is the new arrival that established after peak disturbance, and which might be better adapted to modern climatic conditions that most likely differ from those during the early twentieth century. The lesson seems to be that changing climatic conditions limit ecosystem resilience, preventing recovery to baseline conditions after disturbance as environmental conditions (temperature, precipitation, urbanization, etc.) have also changed. Climate change can also limit the distribution of species, or even exclude them from the regional metacommunity, even in regions that have a great geographic diversity, further hindering ecosystem resilience.

## Conclusions

Lago Verde provides a valuable long-term record of how Neotropical lake ecosystems respond to human disturbance and their subsequent recovery processes. The combination of approaches presented in this research, including remote images, limnology, and paleolimnology, was successful in providing evidence of the long-term evolution of the landscape and documenting different lake responses (resistance, regime-shift, resilience) to disturbance by anthropogenic stressors.

The results show that the lake ecosystem transitioned through three different stages during the last 200 years. Stage 1 represents the baseline condition and included gradual changes under low disturbance. Stage 2 represents a rapid state shift associated with peak disturbance during the 1970s. Stage 3 is an ongoing resilience phase associated with reduced land-use pressure, recent reforestation, and decreasing nitrogen concentrations. Although forest cover has partially recovered and nutrient concentrations have declined, the lake has not returned completely to its original baseline condition; instead, it has reached a new ecological state characterized by partial recovery of baseline taxa, with the incorporation of new species, showing partial long-term resilience.

In a longer time perspective (2000 years), the disturbed condition of the lake during the 1970s and 1980s had a precedent during the Classic period of human occupation of this basin (AD 100 to 800), during which a similar state of the lake was established, attesting that Mesoamerican cultures also had a significant impact on their environment, comparable to modern human-induced changes. Comparing modern conditions with the lake's conditions during

the deforestation event of the Classic period highlights the importance of disturbance duration and climatic context for ecosystem resilience. Past and modern recovery phases indicate that ecosystem resilience is constrained by changing environmental conditions, particularly climate variability. These results suggest that prolonged human disturbance under a changing climate can severely limit ecosystem resilience, making Lago Verde an example relevant to the future of Neotropical aquatic ecosystems in the context of global warming and sustained land-use pressure. These findings emphasize the importance of combining long-term ecological records with modern monitoring to better understand ecosystem trajectories and to inform conservation and restoration strategies in Neotropical regions.

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**Data availability** Data are available at [<https://datosabiertos.unam.mx/IGL:PALEOMEX:VRU-II>] and in Electronic Supplementary Material.

## Declarations

**Conflicts of interest** The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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